

Mapping Mexiletine Binding Sites on the Cardiac Sodium Channel Nav1.5

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The key question, and the short answer

Where does mexiletine bind on Nav1.5, and which site is the best target for drug design? In this study, I ran the molecular-dynamics (MD) simulations and analysed them (stability, where the drug sits, binding energy, pore), then picked a representative structure from each run and used it to search for sites (unbiased (blind) docking, pocket detection, targeted docking), and finally added checks that test whether the new site (from holoMutant-protonated Mex MD simulation) is driven by the mutation. I found two sites. Site 1 is the classic F1760 pore site — the best druggable pocket and the one every study agrees on [2]–[5]. Site 2 is a water-exposed spot at the mouth of the selectivity filter that the protonated drug reached in the mutant, matching the known DIII/IV fenestration region [3],[9].

A few words I use (plain meanings)

Å (ångström): a tiny distance, 0.1 nm; an atom is ~1–2 Å wide. **Docking:** a fast prediction of where a drug fits a structure, based mostly on shape. **MD:** a physics simulation that moves every atom over time. **Occupancy %:** how much of a run the drug stayed in contact with a residue. **Druggability (0–1):** how good a cavity is as a drug pocket (1 = excellent enclosed pocket). **kcal/mol:** a binding-energy unit; more negative = stronger. **Salt bridge:** an attraction between a + and a – charged side chain. **Fenestration:** a side opening between the pore helices, a known drug access route. **Selectivity filter:** the narrow ring at the outer pore that selects sodium ions.

1. Introduction — what I set out to map

What is it? Nav1.5 is the main sodium channel of heart muscle; it opens to let sodium in and start each heartbeat. It is one long protein folded into four domains (DI–DIV) whose four re-entrant loops meet at the outer pore to form the selectivity filter. Mexiletine is a class-Ib sodium-channel blocker used to calm abnormal rhythms. My goal was to map where mexiletine can bind this channel and decide which site is the best drug-design target.

Why did I do it? The binding site is the foundation for designing better drugs and for understanding why some patients respond to a drug and others do not. A concrete example is the N347K mutation in the Domain I P-loop, which nearly abolishes the sodium current and causes Brugada syndrome; experimentally, mexiletine recovers about 21% of the lost current in N347K but barely helps the neighbouring A344S [10]–[12].

What is already known (from the literature)? The best-established site is the **F1760 pore site** on the Domain IV S6 helix: mutating it (F1760A) strongly weakens binding for mexiletine, lidocaine and related blockers [2],[3]. Mexiletine grips the pore with two ends — a chargeable amine (pKa ≈ 9.1, so it is mostly +1 at body pH) and an aromatic ring — making electrostatic and hydrophobic contacts [4].

Beyond this anchor, structural work describes further regions: the central cavity just below the filter (resolved by cryo-EM with the related drug flecainide [5],[7]), the lateral **fenestrations** between pore helices, and a site beneath the intracellular gate [6]; protonated class-Ib drugs can occupy more than one of these, and docking studies place mexiletine in either the inner pore or the DIII/IV fenestration [8],[9]. I modelled both the neutral and the protonated drug, starting from the human Nav1.5 cryo-EM structure (PDB 8VYK [1]).

What does it mean? The literature already paints a **multi-site picture**, with the F1760 pore site as the anchor and the fenestration / DIII–IV region as a secondary route. My task was to reproduce and rank these sites in my own Nav1.5 systems, score them for drug design, and put the binding numbers in context.

2. Methods — how I did it

What is it? Molecular dynamics (MD): a physics simulation that moves every atom in tiny time steps, so I can watch the protein and drug behave over time. Several follow-up tools (docking, pocket detection, binding energy) then read out specific properties from the simulations.

Why did I do it? To map binding sites on equal terms, I built six matched systems and ran each for the same length, so any difference comes from the mutation or the drug — not from different setups.

What did I do? I built six systems (Table 1): WT and N347K, each empty (apo) and with the drug (holo), and the drug neutral or +1 (protonated). Each ran 500 ns in OpenMM (ten 50 ns segments; 10,000 snapshots) in a POPC membrane with water and 0.15 M NaCl (CHARMM36m for protein and lipids; drug parameters from CGenFF). Before each holo simulation run, an initial configuration was established using molecular docking, as simulating the spontaneous diffusion and binding of the drug from bulk solution would be computationally prohibitive within a realistic timeframe. To ensure consistency across all holo systems, the docking search space was standardized using AutoDock Vina [15]: a 26Å x 26Å x 26Å grid box was centered on residue 347 (the mutation site), providing a uniform region to probe. The resulting top-ranked docked pose for each respective system was then utilized as its initial conformation for the MD production runs.

From each finished trajectory I selected a single **representative structure** by DBSCAN clustering of the conformations (the medoid of the most populated cluster; eps 0.53, min_samples 5). These representative structures are what I used for the static, structure-based analyses later — blind docking, pocket detection (fpocket [14]), and targeted docking — so those results describe the most typical shape each system actually adopted, not an arbitrary snapshot. Binding free energies used MM-GBSA (gmx_MMPBSA, igb = 5, 0.15 M, 310 K); the pore profile used HOLE; contacts used a ~4 Å cutoff. I added four checks: a folded-core stability test, through-space distances, an isogeometric charge test (APBS), and motion-correlation (DCCM).

System	Protein	Drug	Drug state
apoWT	Wild-type	—	—
apoMutant	N347K	—	—
holoWT (neutral)	Wild-type	Mexiletine	Neutral
holoMutant (neutral)	N347K	Mexiletine	Neutral
holoWT (protonated)	Wild-type	Mexiletine	+1
holoMutant (protonated)	N347K	Mexiletine	+1

Table 1. The six systems.

3. Results

3.1. System stability (from the MD)

What is it? RMSD is a single number for how far the protein has moved from its starting shape; a related number, RMSF, says how much each residue jiggles. A high RMSD can mean two very different things — the whole fold is breaking down, or just the floppy outer parts are wagging while the core stays put. To separate these, I defined a folded core: the rigid transmembrane body (the four domains, the pore, the selectivity filter), leaving out the large cytoplasmic loops that connect the domains and the mobile N- and C-ends. I built the core objectively, as the 837 of 1395 residues with the lowest RMSF, rather than guessing which loops to drop.

Why did I do it? Every binding-site result that follows sits on this scaffold. If the core were unstable, none of the site mapping, energies, or distances could be trusted; if only the loops move, the results are solid. So this is the foundation check.

What did I find? Whole-protein RMSD was high in every system (6.6–13.1 Å) and especially jumpy in the protonated mutant (13.14 ± 6.61 Å). But the folded core stayed steady everywhere (2.4–2.9 Å), and the contrast is sharpest exactly where it matters: in the protonated mutant the spread collapses from ± 6.61 Å (whole protein) to **± 0.19 Å (core)** — in fact the steadiest core of all six systems (Table 2).

System	Whole-protein RMSD (Å)	Core RMSD (Å)
apoWT	8.12 ± 2.29	2.60 ± 0.25
apoMutant	7.90 ± 0.69	2.86 ± 0.22
holoWT (neutral)	6.58 ± 0.53	2.44 ± 0.17
holoMutant (neutral)	7.91 ± 0.99	2.59 ± 0.25
holoWT (protonated)	8.31 ± 1.02	2.83 ± 0.32
holoMutant (protonated)	13.14 ± 6.61	2.67 ± 0.19

Table 2. Whole-protein vs folded-core backbone RMSD. Core = the 837 of 1395 residues with the lowest RMSF.

What does it mean? The big RMSD numbers are **floppy outer loops, not core collapse**. The transmembrane body that holds the pore and both binding sites is stable in all six runs — including the protonated mutant. So the site mapping that follows rests on a trustworthy scaffold.

3.2. Where the drug settles during the simulation

What is it? I followed the drug through all 10,000 snapshots of each run and recorded which residues it contacted. A residue counts as “touched” in a snapshot when the drug comes within ~4 Å of it (close enough to be a genuine interaction); the occupancy % is how much of the whole run that contact lasted. A high occupancy therefore may mean a stable, long-lived binding pose, not a fleeting brush.

Why did I do it? Docking only guesses a starting pose; MD shows where the drug actually goes and stays once the protein and water are free to move. I wanted to see whether a stable pose forms, and whether it depends on the drug’s charge and on the mutation.

What did I find? The behaviour is split cleanly by charge and by mutation (Table 3). In the protonated mutant, the drug left its docked P-loop start within ~50 ns and settled into a new region I call Site 2 (Domain III/IV), staying there for 95.2% of the run, with its strongest contacts to two acidic residues, ASP1714 (67.9%) and ASP1423 (65.4%), plus greasy contacts to VAL1400 (54.9%) and ILE1424 (25.5%). In the neutral mutant, the same drug stayed up in the Domain I P-loop (PRO348 88.0%, THR353 88.8%, ASP349 74.6%, PHE355 49.8%). In both WT runs, no comparable stable site formed.

System / drug state	Where it settled	Key contacts	Occupancy
holoMutant (protonated)	Site 2 (DIII/IV)	ASP1714, ASP1423, VAL1400	95.2%
holoMutant (neutral)	P-loop (DI)	PRO348, THR353, ASP349	≈88%
holoWT (protonated)	no stable site	—	—
holoWT (neutral)	no stable site	—	—

Table 3. Where mexiletine settled, by mutation and protonation state.

What does it mean? MD simulations revealed a second binding site (Domain III/IV), that the drug accessed exclusively in its protonated state within the mutant system, providing a clear directional insight. However, two important caveats qualify this finding as a strong lead rather than a settled fact: first, this migration event was observed in a single trajectory; second, because all holo simulations originated from the same standardized P-loop region due to the docking setup, the drug's final destination was inherently influenced by its starting domain. Furthermore, the blend of acidic and hydrophobic contacts indicates that Site 2 functions as an electrostatic-plus-hydrophobic pocket—characteristic of the fenestration-type regions described in the literature [8],[9]—rather than a simple charge trap.

3.3. The local K347–D356 salt bridge

What is it? A salt bridge is an attraction between a positively and a negatively charged side chain. The mutation swaps a neutral asparagine (N347) for a positive lysine (K347), and that new + charge can pair with a nearby – residue, D356.

Why did I do it? To pin down what the mutation actually does at the structural level — and, importantly, whether the mutated residue ever touches the drug. This is the groundwork for the later question of whether the mutation can influence Site 2 (Section 3.9).

What did I find? In the mutant, K347 forms a salt bridge with D356, present for 39.2–51.2% of the time depending on the system (Table 4), with a typical K347–D356 distance of ~5.8 Å and a true close contact (< 4 Å) in 44% of frames in the protonated mutant. Critically, the K347 side chain **never contacts the drug (0%)**. WT has neutral N347, which cannot form this salt bridge.

System	K347–D356 salt-bridge occupancy
apoMutant	39.2%
holoMutant (neutral)	51.2%
holoMutant (protonated)	44.0%

Table 4. K347–D356 salt-bridge occupancy in the mutant systems. WT (N347) cannot form it.

What does it mean? The direct structural impact of the mutation is strictly localized, forming an intra-domain salt bridge within Domain I without ever making direct contact with the drug. This structural isolation establishes a critical mechanistic boundary: any influence the mutation exerts on the distant Site 2 must be indirect, mediated through long-range conformational changes. This is precisely the hypothesis investigated and tested in Section 3.9.

3.4. Binding strength, in context (MM-GBSA)

What is it? MM-GBSA estimates how strongly the drug binds, from the MD trajectory (more negative = stronger). These values were evaluated against internal references for consistency and thoroughly cross-referenced with the published literature.

Why did I do it? To put a number on the WT-versus-mutant binding difference and judge whether the magnitude is meaningful — one of the explicit aims of the study.

What did I find? The mutant bound more strongly than WT in all four comparisons (Table 5): protonated –15.01 vs –1.28 kcal/mol (difference –13.73), and neutral –18.47 vs –2.84 (difference –15.63).

System	Drug state	Site	Binding energy (kcal/mol)
holoWT	Neutral	dispersed	–2.84 ± 5.10
holoMutant	Neutral	P-loop	–18.47 ± 3.81
holoWT	Protonated	dispersed	–1.28 ± 3.47
holoMutant	Protonated	Site 2 (Domain III/IV)	–15.01 ± 5.59

Table 5. MM-GBSA binding energies. $\pm$ are within-run frame spreads, not replica errors.

What does it mean? Two honest points. First, the direction (mutant stronger) is consistent across all four runs, but the magnitude is not reliable: the $\pm$ values rival the means, there is no entropy term, and MM-GBSA is at its weakest for a charged drug at an acidic site. Second, on context: the WT value (-1.28 , effectively “no binding”) is not a true WT affinity — mexiletine genuinely binds WT at F1760 [2],[3]; the drug simply never visited that spot in this one WT run — so the meaningful comparison is direction, not magnitude. The literature offers a structural reference rather than an energy: cryo-EM places the related class-Ib drug flecainide in the Nav1.5 central pore cavity [5],[7] — the same pore region that contains my Site 1 (the F1760 local-anaesthetic site). This confirms that the pore site is a physically real drug-binding location, but it gives no comparable binding number, so judging whether -15 kcal/mol is reasonable still needs an orthogonal method (FEP/TI; Section 6).

3.5. Pore width

What is it? HOLE traces the channel pore and measures its narrowest radius (the “bottleneck”). I summarised each 500 ns run by averaging the pore-radius profile over the trajectory and then reading off its narrowest point. This is a purely geometric read-out.

Why did I do it? Not to measure conduction (these runs cannot — see below), but as a structural control: to confirm that neither the drug nor the mutation distorts, closes, or widens the pore, and that the channel keeps its starting open-state geometry across 500 ns. If the pore stayed intact and identical across systems, then the binding results in the other sections sit on a physically stable channel rather than on a structure that the drug or the long simulation has deformed.

What did I find? Per system, the average-profile bottleneck was 1.40 – 1.84 Å — all clustered together (a range of only 0.44 Å) and close to the 1.5 Å sodium size (Table 6). As a stricter cross-check I also took each frame's narrowest point and then averaged; that gives smaller numbers (0.58 – 0.78 Å, because it catches every brief pinch), but the key point is the same: every system-to-system difference (≤ 0.21 Å) was smaller than the frame-to-frame noise (≈ 1.0 Å).

System	Bottleneck (Å)
apoWT	1.56
apoMutant	1.82
holoWT (neutral)	1.66
holoMutant (neutral)	1.40
holoWT (protonated)	1.84
holoMutant (protonated)	1.76

Table 6. Bottleneck of the 500 ns average pore profile, per system.

What does it mean? The pore stays intact and essentially identical across all six systems so neither the drug nor the mutation distorts, opens, or closes it. Two clarifications fix what this number is and is not. First, my starting structure 8VYK is an open state of human Nav1.5 that conducts Na^+ in the original authors' dedicated permeation simulations [1], so the narrow

bottleneck I measure (~ 1.5 Å radius) is the selectivity filter — intrinsically narrow even in a conducting channel — not a closed gate; its z-position on the pore axis also helps place Site 2 at the filter mouth (Section 3.9). Second, and more fundamentally, these are equilibrium simulations with no applied membrane voltage or ion gradient, so they cannot model conduction or current at all: Nav1.5 is voltage-gated, and current is a voltage-driven phenomenon. So HOLE here serves as a geometric integrity check — confirming the pore is stable and undistorted.

From here on: analyses on the representative structures

The next three sections work on the DBSCAN representative structure of each system (Methods) — the most typical shape each run adopted. I let the drug search each structure freely (blind docking), detected and scored every pocket (fpocket) and then docked the drug into the pockets that were found.

3.6. Unbiased site search (blind docking)

What is it? Blind docking scans the entire channel surface without predefined search grids to evaluate unbiased binding preferences across numerous candidate poses.

Why did I do it? To identify the channel's intrinsic hotspots and test which sites the drug naturally prefers without starting bias.

What did I find? A total of 20 poses were generated per system. An analysis restricted to the top-ranked poses showed that protonated mexiletine targeted the canonical F1760 pore site in 3 of 8 receptors, scattered across miscellaneous surface spots in the rest, and reached the Domain III/IV region (Site 2) in none of them.

What does it mean? Based on the top-scoring static poses, the drug defaults to the F1760 pore site—the literature's canonical region—rather than Site 2. This creates a critical contrast with the MD data: Site 1 acts as a geometric default, whereas Site 2 requires the temporal coordination of protein dynamics, water, and electrostatics to emerge (Section 3.2), indicating that these two sites are driven by fundamentally different physical principles.

3.7. Pocket detection and druggability (fpocket)

What is it? Think of fpocket as a digital scanner that looks over the protein's surface to find hidden cavities or "cups" where a drug might sit. It grades every cavity with a score from 0 to 1. A score close to 1 means it is an excellent, well-enclosed pocket—it is just the right size and has greasy (hydrophobic) walls that help "glue" the drug in place. A score near 0 means the surface is too flat or open, making it a poor place for a drug to bind. To be absolutely sure about the main pocket (the classic F1760 site), I didn't just check it once; I checked it across 60 different moving snapshots of the protein to see if its good score holds up over time.

Why did I do it? I had two clear goals:

To make a list of available pockets: I scanned the main structure of each system to see what kind of pockets exist, specifically tracking down the classic **F1760 pore pocket (Site 1)** and the **Domain III/IV junction pocket (Site 2)**. This list tells us exactly where to try and plug the drug in during the next step.

To test if the pocket is reliably stable: I wanted to make sure the F1760 pocket stays open and usable during normal protein movements, rather than just looking good in one lucky, accidental frame.

What did I find? In the Static Structures (No movement): In the raw, static WT structure directly from the PDB (8VYK), and in the static Mutant structure where I introduced the N347K mutation directly inside PyMOL, the F1760 pocket scored exactly **0.688** for both. This shows that without protein movement, simply swapping a residue in PyMOL has zero geometric effect on this distant pocket.

In the MD Representative Structures (With movement): Once the protein was free to move in the simulations, the results changed significantly. In the DBSCAN representative structures—which capture the most typical shapes the protein actually adopted—the F1760 pocket scored much higher, jumping to **≈ 0.81–1.00** (for example, it is Pocket 7 with a score of 0.991 in the protonated mutant). This confirms that simulation dynamics help reveal a beautifully well-formed drug pocket.

Across the 60-Frame Sampling: Looking closely at the 60 snapshots per system showed that this pocket is highly flexible. It constantly breathes, swinging between perfectly open (~1.0) and temporarily collapsed (~0.0). Still, it remains highly usable for a drug in most frames, with averages between 0.58 and 0.82, and is druggable (≥ 0.7) 57% to 78% of the time.

Site 2 (Domain III/IV): In sharp contrast, this region scored practically zero (**0.010 to 0.012**) even in the moving representative structures, meaning it does not look like a geometric drug pocket at all.

System	F1760 druggability (mean $\pm$ SD)	Range	Frames ≥ 0.7
apoWT	0.58 $\pm$ 0.42	0.00–1.00	57%
apoMutant	0.70 $\pm$ 0.38	0.00–1.00	68%
holoWT (neutral)	0.73 $\pm$ 0.38	0.00–1.00	73%
holoMutant (neutral)	0.82 $\pm$ 0.29	0.04–1.00	78%
holoWT (protonated)	0.68 $\pm$ 0.39	0.00–1.00	68%
holoMutant (protonated)	0.77 $\pm$ 0.37	0.00–1.00	78%

Table 7. F1760-pocket druggability over 60 frames per system. Static: 0.688 (identical WT vs mutant). Site-2 region: 0.010–0.012 (non-druggable).

What does it mean? This tells us a very clear story about how the two sites work:

Site 1 (F1760) is hands-down the best target for designing a drug. It is a real, reliable, enclosed pocket. Highlighting the difference between the rigid PyMOL models and the moving MD structures proves why static scoring isn't enough: the single "representative" structure catches the pocket at an exceptionally high state (~1.0), but in reality, the pocket opens and closes like a valve over time, making the honest score a distribution rather than a single number.

Site 2's score of nearly 0 does *not* mean the software missed it; it means the software described it perfectly. Site 2 isn't a cozy, enclosed pocket—it is an open, water-exposed mouth. Because computer programs that look purely at *shape* (like fpocket or blind docking) look for an enclosed cup, they completely dismiss Site 2. However, physics simulations (MD) still find the drug there because Site 2 acts like a magnet, pulling the drug in using electrical charges (electrostatics) rather than a snug physical fit.

3.8. Site-specific docking into the detected pockets

What is it? Site-specific (or targeted) docking is an approach where you restrict the software's search space to a specific, predefined region of the protein. Instead of letting the drug search the entire channel, you focus the grid box on a single candidate pocket. This allows us to score and compare how well the drug fits into different sites using the exact same scoring scale. It is the focused counterpart to the unbiased "blind docking" from Section 3.6.

Why did I do it? During the blind docking run, the drug naturally favored the F1760 pore and never sampled Site 2. Because blind docking completely bypassed Site 2, it couldn't generate a binding score for that region. To evaluate both sites fairly and give them an equal baseline, I performed site-specific docking directly into the coordinates of both pockets identified by fpocket—**Site 1 (F1760)** and **Site 2 (Domain III/IV)**—across every system where they were detected.

What did I do? I used the exact same DBSCAN representative structures from the blind docking analysis, meaning the receptor conformation remained identical. The only variable that changed was the search space. For each target site, I centered a fine-tuned grid box on the exact pocket coordinates reported by fpocket.

What did I find? Comparable Local Scores: Interestingly, when the search was confined to these local spaces, both sites yielded binding scores in the exact same range, from **-5.2 to -6.3 kcal/mol**.

Site 2 Scores Well Locally: Even though fpocket assigned a near-zero geometric druggability score to Site 2, focusing the search space on this region allowed the drug to find a favorable local pose scoring -6 kcal/mol.

Conserved Interactions: In this site-specific pose, the protonated (+1) drug formed tight, close contacts with two key acidic residues: **ASP1714** (2.11 Å) and **ASP1423** (3.41 Å). These are the exact same two residues that the drug is bound to during the long molecular dynamics (MD) simulations.

What does it mean? This comparison reveals a critical methodological insight regarding the limitations of purely geometry-based software:

The Catch of Local Search Spaces: While both sites look comparable on a local shape-and-contact scale, Site 2 only achieves a good score when the software is explicitly directed to search there. When left entirely unbiased (as seen in blind docking), the drug never selects Site 2 on its own.

Why Geometric Tools Miss It: Both fpocket (with its 0.010 score) and blind docking heavily underrate Site 2 because they evaluate binding through a geometric lens. They see an open, water-exposed surface and dismiss it as a poor pocket. However, Site 2's driving force is electrostatic, acting like a chemical magnet where the positively charged drug is pulled in by the negative fields of ASP1714 and ASP1423.

The Verdict: Only physics-based tools (MD + MM-GBSA) can fully capture this charge-driven, surface-exposed binding. Ultimately, **Site 1 (F1760)** stands out as the reliable, self-selecting drug target. **Site 2** is a valid electrostatic interaction site, but it is a secondary region that geometric tools alone are blind to.

3.9. What is Site 2, and is it driven by the mutation?

What is it? This section answers two connected questions: What part of the channel is Site 2 exactly? And, since the drug only visited Site 2 in the mutant simulation, did the N347K mutation actually cause this?

Why did I do it? I wanted to map Site 2 onto the known structure of the channel and test the natural guess that the mutation was responsible. To find out, I looked at physical distances, performed an electrical charge test, and checked if the movements of the two regions were linked (motion correlation).

What did I find? (a) Site 2 is the selectivity-filter mouth: Site 2 is simply the outer entrance of the channel's selectivity filter. Here, two negative amino acids, ASP1423 and ASP1714, form a natural, negatively charged ring designed to pull in positive ions. This perfectly explains why our positively charged (protonated) drug is pulled there, and it perfectly matches the "DIII/IV fenestration" region already known in the literature.

(b) The mutation does NOT create it: Residue 347 is roughly 26Å away from Site 2 in every single system (never getting closer than 12Å). Because that distance is very far, I tested the only two ways a mutation could still influence a distant site:

(H1) The Electrical Charge Effect: I ran a test switching the positive charge of K347 on and off. Its electrical contribution at Site 2 was practically zero (+0.032 kT/e) and actually had the wrong sign (a positive charge makes a distant spot less negative, meaning it would push the positive drug away, not attract it).

(H2) Long-Range Movement Link (Motion Coupling): The movement correlation between residue 347 and Site 2 was very low and virtually identical between the mutant (0.137) and the wild-type (0.152). As a control, residue 347 showed a very strong connection to its nearby partner D356 (+0.30 to +0.74), proving the measurement method works perfectly.

What does it mean? Site 2 is a real, well-known part of the channel (the filter mouth), but it sits 26Å away from the mutation. Since neither electrical forces (H1) nor protein movements (H2) connect them, the mutation does not create Site 2. Instead, Site 2 is just a general landing spot that any protonated drug can visit. The fact that the drug settled there only in the mutant run is likely a coincidence of that single simulation trajectory, and repeating the simulations (Section 6) is required to know for sure.

4. Best site for drug design, and a map of all sites

4.1. Ranking the two sites

What did I find? Pulling the four mapping methods together, Site 1 is the reliable drug-design target and Site 2 is a secondary, open electrostatic site (Table 8).

Question	Site 1 (F1760 pore)	Site 2 (filter mouth)
Chosen by blind docking?	Yes (3/8)	No
A real druggable pocket?	Yes (0.58–0.82)	No (0.010–0.012)
Seen in MD?	Default site in the literature	95.2% (protonated, one run)
Targeted-docking score (after fpocket)	–5.2 to –6.3	–6.0 (hand-placed)
Supported by literature?	Strongly [2]–[5]	Fenestration region [3],[9]
Mutation-specific?	No	Likely, no (~26 Å away)
Drug-design verdict	Primary target	Secondary / electrostatic

Table 8. The two sites I mapped, ranked across all methods.

4.2. Where these sit on the known map of Nav drug sites

What did I find? The two sites fit neatly into the binding regions already reported for mexiletine and related blockers (Table 9): Site 1 is the canonical pore site, Site 2 is the fenestration / DIII–IV region.

Drug (study)	Site / region	How found	My match
Mexiletine [2]	F1760 pore (DIV-S6)	Mutagenesis + ephys	Site 1
Mexiletine / BPA [3]	F1760 + DIII–IV fenestration	F1760A + steered MD	Site 1 + Site 2 route
Mexiletine [9]	inner pore OR III/IV fenestration	docking into cryo-EM	two-site model
Flecainide [5],[7]	central cavity below filter	cryo-EM (6UZ3) + MD	near Site 1
Lacosamide/mexiletine [8]	fenestration + central pore	cryo-EM + review	Site 2 region
Carbamazepine [6]	beneath intracellular gate	cryo-EM (Nav1.7)	not seen here

Table 9. My two sites mapped onto the binding regions reported in the literature.

What does it mean? The map is consistent: **Site 1 (F1760) is the established, druggable pore site and the right primary target**; Site 2 corresponds to the known fenestration / selectivity-filter-mouth region that protonated class-Ib drugs are known to visit. A third literature site (beneath the intracellular gate) was not sampled here and is a clear target for future work.

5. Conclusions & Discussion

- Mexiletine has two binding regions on my Nav1.5 systems: Site 1 (F1760 pore) and Site 2 (selectivity-filter mouth / fenestration region).
- Site 1 is the best drug-design target: unbiased docking chooses it (3/8), it is a genuine druggable pocket (0.58–0.82), and the literature strongly supports it [2]–[5].
- Site 2 is a real but open, water-exposed electrostatic site reached only by the protonated drug in MD (95.2% in one run); it matches the known DIII/IV fenestration region [3],[9] but is not a closed pocket (0.010–0.012).
- MM-GBSA shows the mutant binding more strongly ($\Delta\Delta G$ –13.73 protonated, –15.63 neutral), a consistent direction — but single-run, entropy-free and noisy, so not an established magnitude.
- The structure is stable at its core (2.4–2.9 Å), and pore widths do not differ between systems (within noise), so these results map binding rather than conductance.
- Side question — is Site 2 caused by N347K? Not by any mechanism I could find: the mutation is ~26 Å away, its charge contributes only +0.032 kT/e there, and there is no excess motion-coupling (0.137 vs 0.152).

A Quick Discussion: Real Selection or an Electrostatic Escape?

There is an important question we must ask about why the drug moved to Site 2 so quickly (~50 ns) in the mutant run. Did the drug actively choose this site, or could this movement be an artifact of how the simulation started? Because letting a drug swim freely in water from the start takes too much computer time, all setups began with the drug placed inside the Domain I P-loop, right next to residue 347.

In the protonated mutant system, this setup placed a positively charged drug (+1) immediately next to a mutation that adds a new positive charge (K347). It is highly possible that as soon as the simulation started, these two identical positive charges repelled each other like matching magnets.

Therefore, instead of "choosing" Site 2 because of a true mutation-driven preference, the drug might simply have been kicked out of the P-loop by electrical repulsion. Once pushed into the local water, it would naturally fall into the nearest available negative sink—the selectivity filter mouth (Site 2). To test if this was just a forced escape rather than a true preference, we would need the future simulations proposed in Section 6: either letting the drug enter freely from bulk water, or starting the MD from the initial configurations selected by our unbiased blind docking.

6. Future works open to discussion

- Repeat the runs ($\geq 3 \times 500$ ns) to test whether the Site-2 visit and 95.2% occupancy reproduce, and to turn within-run spreads into real error bars.
- A better binding-energy method (FEP/TI), to test whether the $\Delta\Delta G$ survives a rigorous, entropy-aware calculation.
- Let the drug enter freely from bulk water (unbiased), or initiate the MD simulations from the alternative configurations selected by our unbiased blind docking; this will remove the shared P-loop starting bias and reveal whether the drug can still find Site 2 entirely on its own.
- Extend to A344S and to the intracellular-gate site, to complete the binding-site map and connect binding to the experimental current rescue [10],[11].

HPC Resource Requirements and Project Timeline

To realistically execute these proposed future studies, additional dedicated HPC (High-Performance Computing) server support is required; completing these runs with my current resources alone would simply take too much time.

Based on our current benchmarks, **Server A** completes one 500 ns production run in 10 days, while **Server B** completes the same run in 17 days. For the 32 planned simulation runs (8 replication runs, 12 bulk-water entry or blind docking initial configuration simulations, and 12 A344S comparison runs), we will implement a continuous greedy dispatch strategy—instantly sending a simulation to whichever server becomes free first. Under this continuous dispatch plan, Server A will handle 20 runs and Server B will handle 12 runs, ensuring neither server idles materially. The entire workload will be completed in approximately **7 months**.

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Appendix

Repository

All trajectories, analyses, figures, docking results and scripts:
github.com/yagmur-yesilyurt/bioinformatics_master.